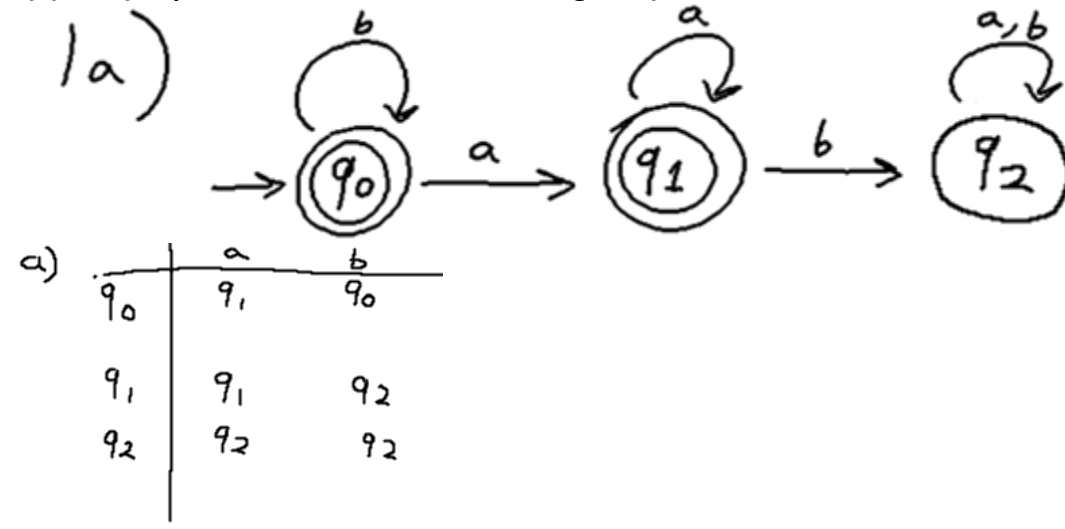


Homework 2

Name: Eristen Forsythe

1(a) $L = \{ w \mid w \text{ does NOT contain substring 'ab'} \}$



States:

q0 = start

q1 = just saw 'a'

q2 = seen 'ab'

Transitions:

q0 --a--> q1

q0 --b--> q0

q1 --a--> q1

q1 --b--> q2

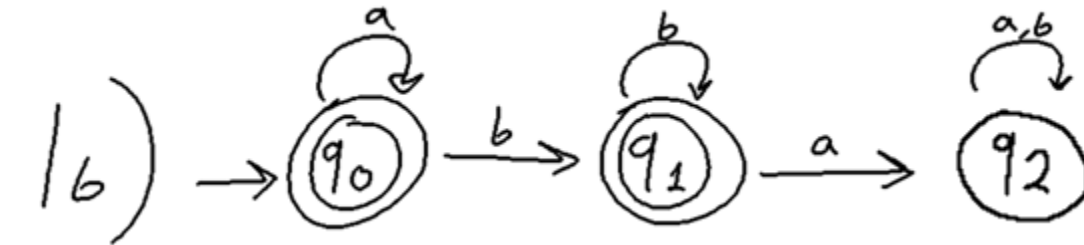
q2 --a,b--> q2

Final accepting states: {q0, q1}

Homework 2

Name: Eristen Forsythe

1(b) $L = \{ w \mid w \text{ does NOT contain substring 'ba'} \}$



b)

	a	b
q ₀	q ₀	q ₁
q ₁	q ₂	q ₁
q ₂	q ₂	q ₂

States:

q₀ = start

q₁ = just saw 'b'

q₂ = seen 'ba'

Transitions:

q₀ --b--> q₁

q₀ --a--> q₀

q₁ --b--> q₁

q₁ --a--> q₂

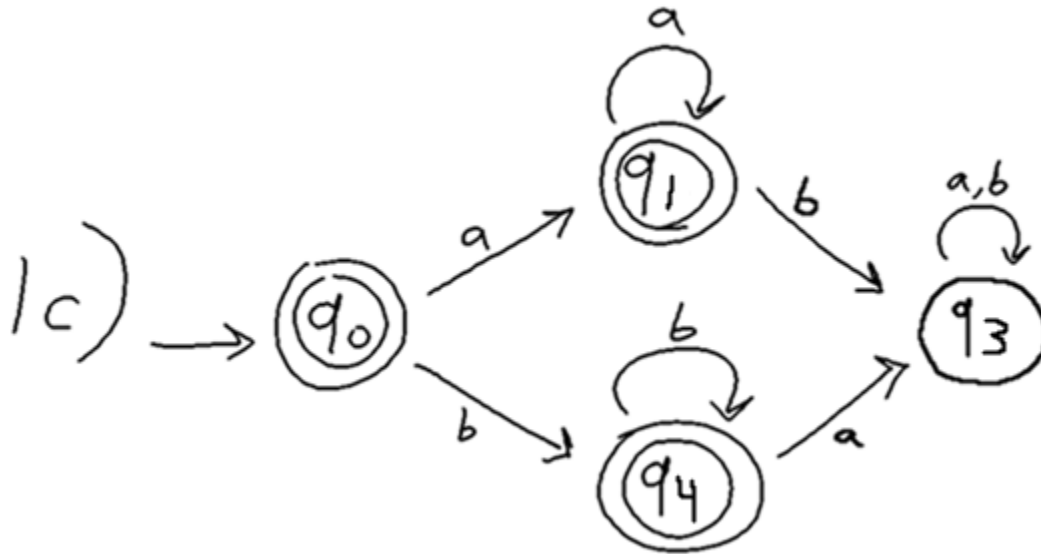
q₂ --a,b--> q₂

Final accepting states: {q₀, q₁}

Homework 2

Name: Eristen Forsythe

1(c) $L = \{ w \mid w \text{ contains neither 'ab' nor 'ba' } \}$



c)

	a	b
q ₀	q ₁	q ₂
q ₁	q ₁	q ₃
q ₂	q ₃	q ₂
q ₃	q ₃	q ₃

States:

q₀ = start

q₁ = seen only a's

q₂ = seen only b's

q₃ = dead state

Transitions:

q₀ --a--> q₁

q₀ --b--> q₂

q₁ --a--> q₁

q₁ --b--> q₃

q₂ --b--> q₂

q₂ --a--> q₃

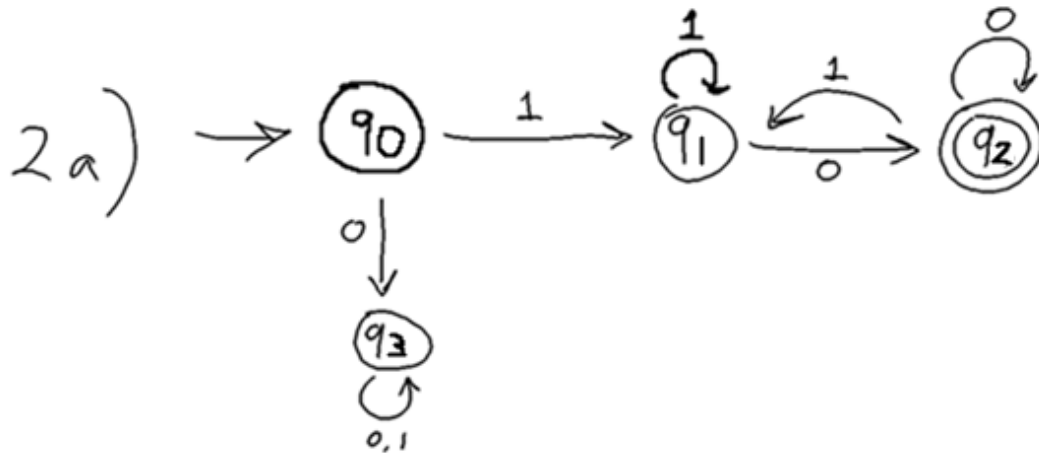
q₃ --a,b--> q₃

Accepting states: {q₀, q₁, q₂}

Homework 2

Name: Eristen Forsythe

2(a) $L = \{ w \mid w \text{ begins with 1 and ends with 0} \}$



a)

	0	1
q ₀	q ₃	q ₁
q ₁	q ₂	q ₁
q ₂	q ₂	q ₁
q ₃	q ₃	q ₃

States:

Q0 = start

Q3 = dead state

q1 = started with 1, last digit 1

q2 = started with 1, last digit 0

Transitions:

Q0 --1--> q1

Q0 --0--> q3

Q3 --0,1--> q3

q1 --1--> q1

q1 --0--> q2

q2 --0--> q2

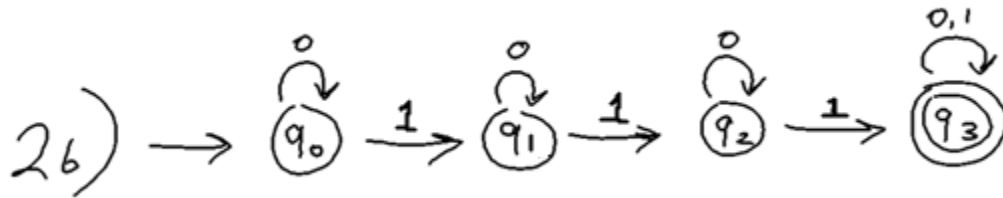
q2 --1--> q1

Accepting State: {q2}

Homework 2

Name: Eristen Forsythe

2(b) $L = \{ w \mid w \text{ contains at least three 1s} \}$



b)

	0	1
q ₀	q ₀	q ₁
q ₁	q ₁	q ₂
q ₂	q ₂	q ₃
q ₃	q ₃	q ₃

States:

q₀ = 0 ones

q₁ = 1 one

q₂ = 2 ones

q₃ = 3 or more ones

Transitions:

q₀ --1--> q₁ ; q₀ --0--> q₀

q₁ --1--> q₂ ; q₁ --0--> q₁

q₂ --1--> q₃ ; q₂ --0--> q₂

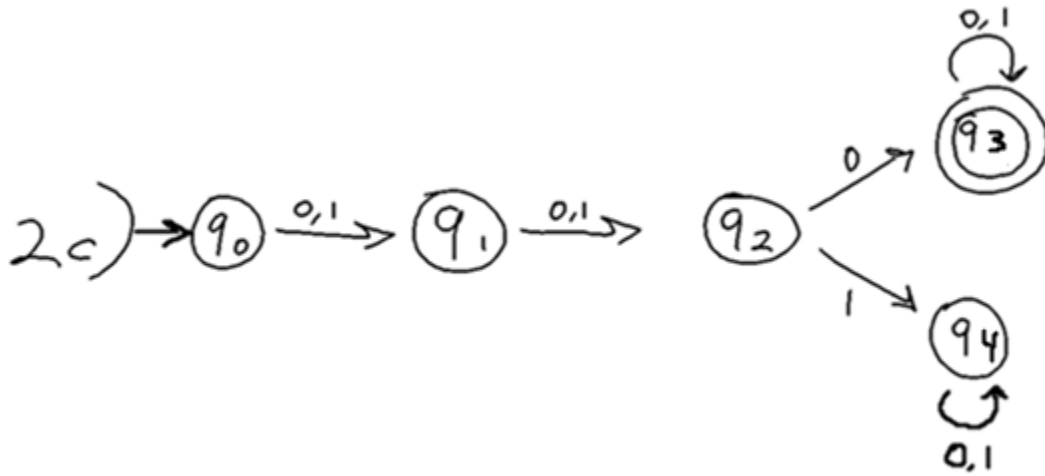
q₃ --0,1--> q₃

Accepting State: {q₃}

Homework 2

Name: Eristen Forsythe

2(c) $L = \{ w \mid \text{length} \geq 3 \text{ and third symbol is } 0 \}$



c)

	0	1
q ₀	q ₁	q ₁
q ₁	q ₂	q ₂
q ₂	q ₃	q ₄
q ₃	q ₃	q ₃
q ₄	q ₄	q ₄

States:

q0 = start

q1 = read 1 symbol

q2 = read 2 symbols

q3 = third symbol was 0

q4 = third symbol was 1

Transitions:

q0 --0,1--> q1

q1 --0,1--> q2

q2 --0--> q3

q2 --1--> q4

q3 --0,1--> q3

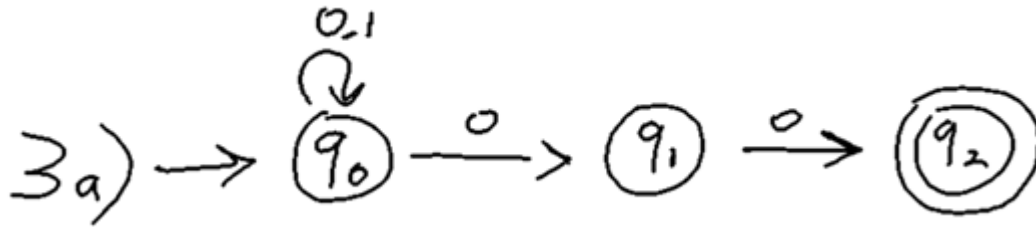
q4 --0,1--> q4

Accepting State: {q3}

Homework 2

Name: Eristen Forsythe

3(a) NFA for $L = \{ w \mid w \text{ ends with } 00 \}$ (3 states)



States:

q_0 = start

q_1

q_2

Transitions:

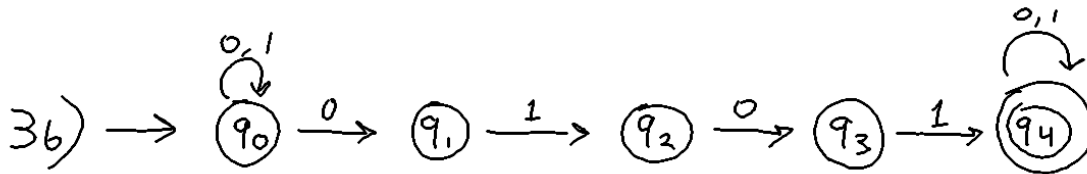
$q_0 \xrightarrow{0} q_0$ and q_1

$q_0 \xrightarrow{1} q_0$

$q_1 \xrightarrow{0} q_2$

Accepting State: $\{q_2\}$

3(b) NFA for $L = \{ w \mid w \text{ contains substring } 0101 \}$ (5 states)



States:

q_0

q_1

q_2

q_3

q_4

Transitions:

$q_0 \xrightarrow{0} q_0$ and q_1

$q_0 \xrightarrow{1} q_0$

$q_1 \xrightarrow{1} q_2$

$q_2 \xrightarrow{0} q_3$

$q_3 \xrightarrow{1} q_4$

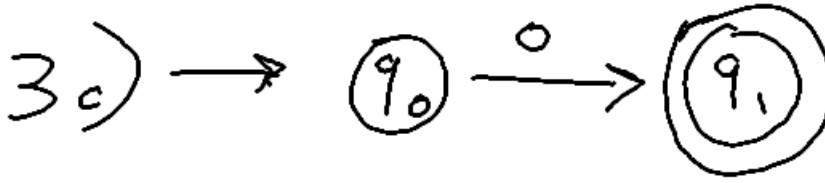
$q_4 \xrightarrow{0,1} q_4$

Accepting State: $\{q_4\}$

Homework 2

Name: Eristen Forsythe

3(c) NFA for $L = \{0\}$ (2 states)



States:

q_0 = start

q_1

Transition:

$q_0 \xrightarrow{0} q_1$

Accepting State: $\{q_1\}$